DESIGNING MANUFACTURABLE PRODUCTS By Simplifying Complexity

Building upon the author's previous article on product risk analysis, this article delves into the often overlooked link between design and manufacturing, where poor design choices can lead to costly issues. The article explores how measuring process complexity and reducing intricacies can empower the product development process

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In 1957, The Michigan Technique—a renowned magazine focussing on machine tools and production techniques—published a poem that humorously depicted the ongoing struggle between designers and the shop floor. The poem highlighted the changes made by the designer and concluded with the lines:

"And it can't be planned and it can't be ground
So I feel my design is unusually sound."
And he shouted in glee,
"Success at last!
This goddam thing can't even be cast."

Unfortunately, this satirical portrayal reflects a reality that we often overlook. Countless examples surround us, demonstrating how poor design has resulted in costly manufacturing and maintenance issues. With a few clicks of a mouse, a design can be altered. However, once the manufacturing stage begins, any modifications to the product design becomes very costly.

Extensive research has revealed that we commit about 63% of the cost at the time of initial design conception. By the time the design is finalised, approximately 85% of the costs have already been committed (Fig. 1). Subsequent processes of manufacturing and distribution offer little room for improvement.

During the design phase, it is crucial to consider manufacturing operations. There are two main aspects of manufacturing problems. One type of problem arises from manufacturing complexity. These complexities may emerge due to different types of components required for creating the parts.

Larger varieties require more inventory holding and more effort in supply chain management. We can also have complexities from a large number of different processes required to make the product.

Process complexities can arise from the diverse range of machines, storage, handling systems, and operators involved. Managing and maintaining the various storage and handling systems can pose challenges and the training required to operate them increases accordingly.

If we do not train the people properly on using different tools and equipment,